MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 3 02/09/2021

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We will discuss several questions from Chapter 4 in Ross' book.

Tutorial Questions

(1) Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:
 (a) the maximum value to appear in the two rolls;
 (b) the minimum value to appear in the two rolls;
 (c) the sum of the two rolls;
 (d) the value of the first roll minus the value of the second roll?

(2) A salesman has scheduled two appointments to sell encyclopedias. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs \$1000, or the standard model, which costs \$500. Determine the probability mass function of X, the total dollar value of all sales.

(3) Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0 \\ \frac{b}{4}, & 0 \leq b < 1 \\ \frac{1}{2} + \frac{b-1}{4}, & 1 \leq b < 2 \\ \frac{11}{12}, & 2 \leq b < 3 \\ 1, & 3 \leq b \end{cases}$$

 (a) Find $P\{X = i\}$, $i = 1, 2, 3$.
 (b) Find $P\{\frac{1}{2} < X < \frac{3}{2}\}$.

(4) Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p. Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show that in both cases that this number is maximized when $p = 1/2$.

(5) One of the numbers 1 through 10 is randomly chosen. You are to try to guess the number chosen by asking questions with yes-no answers. Compute the expected number of questions you will need to ask in each of the following two cases:
 (a) Your i-th question is to be "Is it i?", $i = 1, 2, \ldots, 10$.
 (b) With each question you try to eliminate one half of the remaining numbers, as nearly as possible.

(6) If $\mathbb{E}(X) = 1$ and $\mathrm{Var}(X) = 5$, find
 (a) $\mathbb{E}((2 + X)^2)$;
 (b) $\mathrm{Var}(4 + 3X)$.
 (We assume that X is a discrete random variable.)

(7) Suppose that, in flight, airplane engines will fail with probability $1 - p$, independently from engine to engine. If an airplane needs a majority of its engines operative to complete a successful flight, for what values of p is a 5-engine plane preferable to a 3-engine plane?

(8) It is known that diskettes produced by a certain company will be defective with probability 0.01, independently of each other. The company sells the diskettes in packages of size 10 and offers a money-back guarantee that at most 1 of the 10 diskettes in the package will be defective. The guarantee is that the customer can return the entire package of diskettes if he or she finds more than one defective diskette in it. If someone buys 3 packages, what is the probability that he or she will return exactly 1 of them?

Assignment Questions

(1) Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X? For $n = 3$, find the probabilities associated with the values that X can take on? (We assume the coin is assumed fair.)

(2) In the game of Two-Finger Morra, 2 players show 1 or 2 fingers and simultaneously guess the number of fingers their opponent will show. If only one of the players guesses correctly, he wins an amount by him and his opponent. If both players guess correctly or if neither guesses correctly, then no money is exchanged. Consider a specified player, and denote by X the amount of money he wins in a single game of Two-Finger Morra.
  (a) If each player acts independently of the other, and if each player makes his choice of the number of fingers he will hold up and the number he will guess that his opponent will hold up in such a way that each of the 4 possibilities is equally likely, what are the possible values of X and what are their associated probabilities?
  (b) Suppose that each player acts independently of the other. If each player decides to hold up the same number of fingers that he guesses his opponent will hold up, and if each player is equally likely to hold up 1 or 2 fingers, what are the possible values of X and their associated probabilities?

(3) Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $P\{X = i\}$, $i = 0, 1, 2, 3, 4$.

(4) A gambling book recommends the following "winning strategy" for the game of roulette: Bet \$1 on red. If red appears (which has probability $\frac{18}{38}$), then take the \$1 profit and quit. If red does not appear and you lose this bet (which has probability $\frac{20}{38}$ of occurring), make additional \$1 bets on red on each of the next two spins of the roulette wheel and then quit. Let X denote your winnings when you quit.
  (a) Find $P\{X > 0\}$.
  (b) Are you convinced that the strategy is indeed a "winning" strategy? Explain your answer.
  (c) Find $\mathbb{E}(X)$.

(5) Four buses carrying 148 students from the same school arrive at a football stadium. The buses carry, respectively, 40, 33, 25, and 50 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on her bus.
  (a) Which of $\mathbb{E}(X)$ or $\mathbb{E}(Y)$ do you think is larger? Why?
  (b) Compute $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

(6) Suppose that two teams play a series of games that ends when one of them has won i games. Suppose that each game played is, independently, won by team A with probability p. Find the expected number of games that are played when (a) $i = 2$ and (b) $i = 3$. Also, show in both cases that this number is maximized when $p = 12$.

(7) You have \$1000, and a certain commodity presently sells for \$2 per ounce. Suppose that after one week the commodity will sell for either \$1 or \$4 an ounce, with these two possibilities being equally likely.

(a) If your objective is to maximize the expected amount of money that you possess at the end of the week, what strategy should you employ?

(b) If your objective is to maximize the expected amount of the commodity that you possess at the end of the week, what strategy should you employ?

(8) Each night different meteorologists give us the probability that it will rain the next day. To judge how well these people predict, we will score each of them as follows: If a meteorologist says that it will rain with probability p, then he or she will receive a score of

$$\begin{cases} 1-(1-p)^2 & \text{if it does rain;} \\ 1-p^2 & \text{if it does not rain.} \end{cases}$$

We will then keep track of scores over a certain time span and conclude that the meteorologist with the highest average score is the best predictor of weather. Suppose now that a given meteorologist is aware of our scoring mechanism and wants to maximize his or her expected score. If this person truly believes that it will rain tomorrow with probability p., what value of p should he or she assert so as to maximize the expected score?